

大模型RAG进阶实战营 – 案例篇

01 自己动手制作RAG框架

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Load File

Chunk File

Parse File

Embedding File

Indexing with Vector DB

Similarity Search

Generation

Similarity Search

Your Question

到底什么是大模型蒸馏，
DeepSeek是怎么蒸馏Qwen的？ |

Vector Database

Milvus

Collection

Choose a collection...

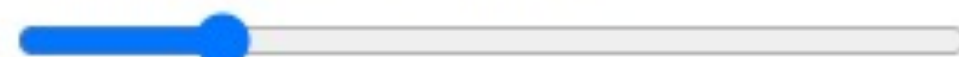
Top K Results

3

Similarity Threshold: 0.7



Minimum Word Count: 100



☐ Save Search Results

Search



Search results will appear here

下载并运行框架

下载框架Repo

黄佳

huangjia2019

Always Studying

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<https://github.com/huangjia2019/rag-project01-framework>

`git clone https://github.com/huangjia2019/rag-project01-framework.git`

Jupyter Notebook Updated 5 days ago

rag-project01-framework Public

这是我用Cursor自行制作的RAG Framework，好处是可以掌控RAG每一个环节的教学细节

RAG-framework Public

master

1 Branch 0 Tags

Go to

huangjia2019 key

backend 20250220

frontend 20250220

.gitignore 祝大家2025学习进步

requirements.txt 祝大家2025学习进步

安装Node.js (Ubuntu)

1 检查本地是否安装了 npm

运行 `npm -v` 命令

```
huangj2@IHP-SPD-CF00015:~/Documents/RAG_P1_自制框架_250310$ npm -v
10.9.2
```

2 安装 npm

```
sudo apt update
sudo apt install nodejs npm -y
```

3 安装前端组件

1. 先进入 `frontend` 目录
2. 运行 `npm install` 命令

```
(ragprojects_250310) C:\Users\huangj2\OneDrive - A STAR\99 Books\2025-极客时间RAG训练营\代码\RAG-framework\frontend>npm install
```

```
added 343 packages, and audited 344 packages in 13s
```

```
127 packages are looking for funding
  run `npm fund` for details
```

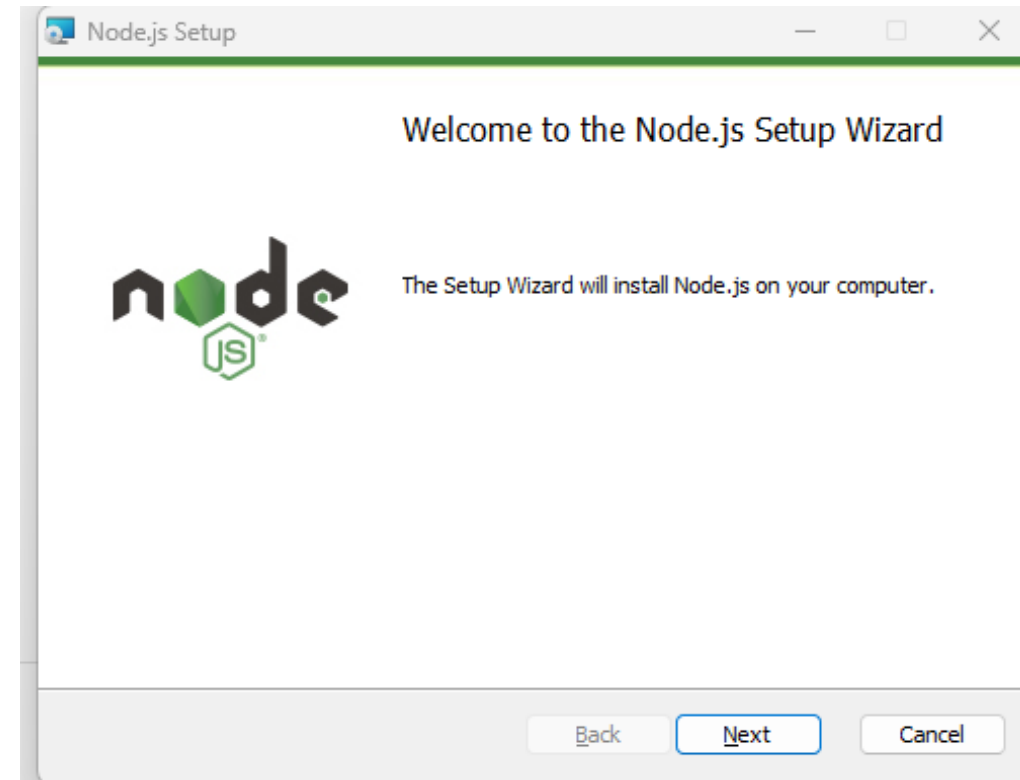
```
3 moderate severity vulnerabilities
```

```
To address all issues, run:
  npm audit fix
```

```
Run `npm audit` for details.
```

安装Node.js (Windows、macOS)

在 <https://nodejs.org/zh-cn/download> 下载安装包然后安装即可



1. 安装完成后进入frontend目录
2. 然后执行 **npm install** 命令

```
(ragprojects_250310) C:\Users\huangj2\OneDrive - A STAR\99 Books\2025-极客时间RAG训练营\代码\RAG-framework\frontend>npm install
```

```
added 343 packages, and audited 344 packages in 13s
```

```
127 packages are looking for funding
  run `npm fund` for details
```

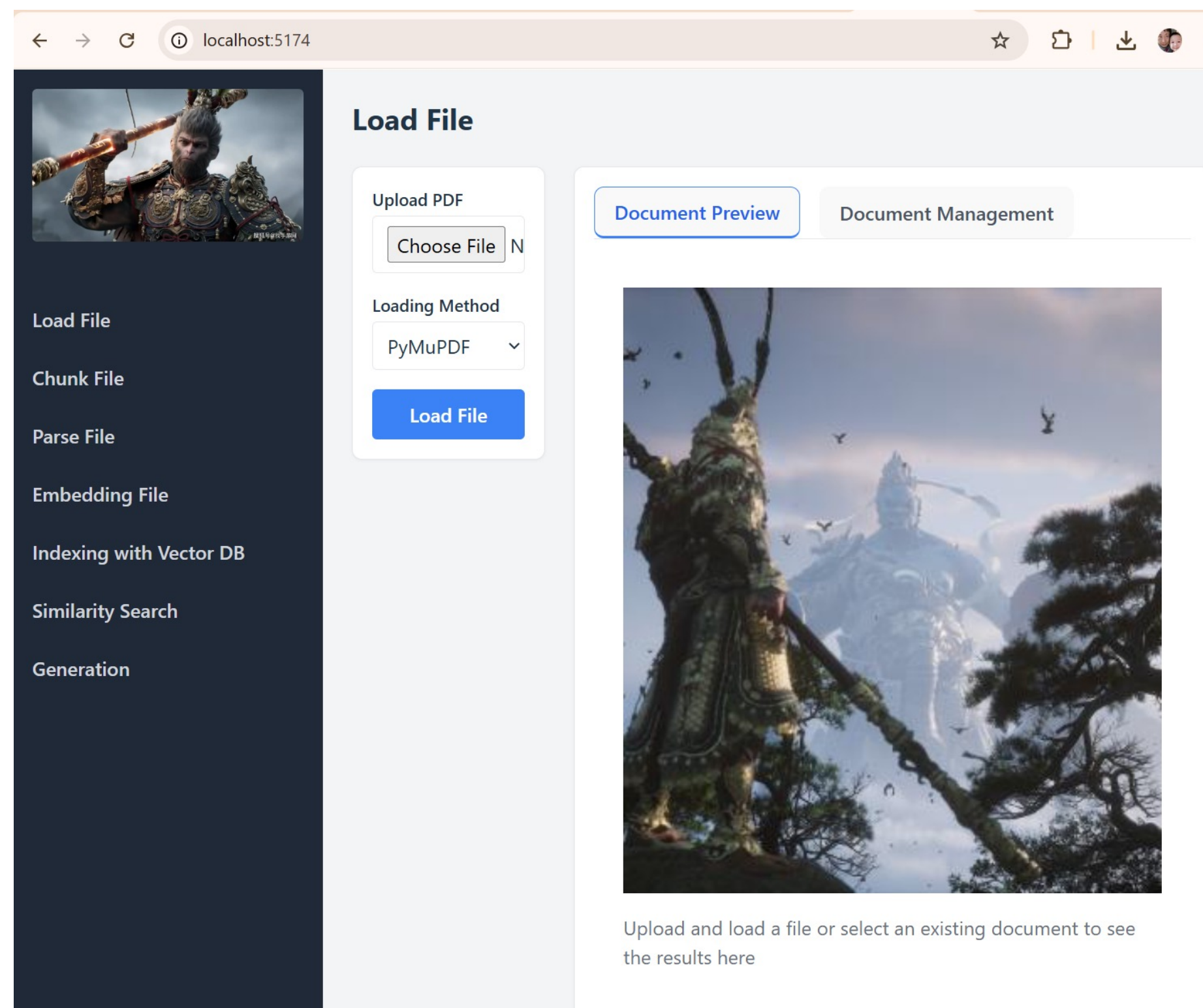
```
3 moderate severity vulnerabilities
```

```
To address all issues, run:
  npm audit fix
```

```
Run `npm audit` for details.
```


运行前端

在 **frontend** 目录下运行 **npm run dev** 命令



Problems Output Debug Console **Terminal** Ports

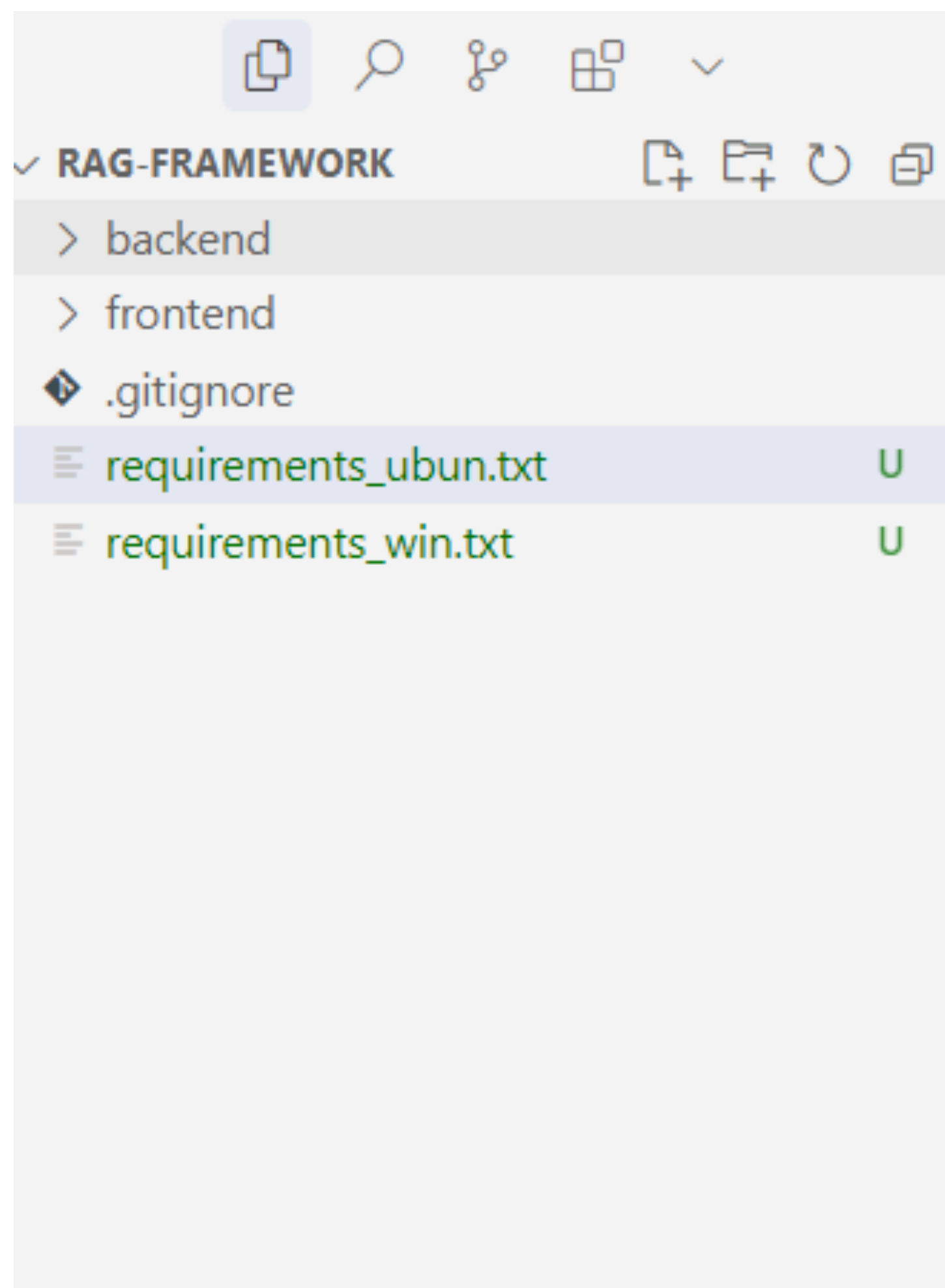
VITE v5.4.11 ready in 224 ms

→ **Local:** http://localhost:5174/
→ **Network:** http://10.2.82.80:5174/
→ press **h** + **enter** to show help



安装环境（后端）

1. `python -m venv Project01`
2. `source .../bin/activate`
3. `pip install -r requirements.txt`



requirements_win.txt U

requirements_ubuntu.txt U X

requirements_ubuntu.txt

```
1 accelerate==1.3.0
2 aiohappyeyeballs==2.4.0
3 aiohttp==3.10.5
4 aiosignal==1.3.1
5 annotated-types==0.7.0
6 antlr4-python3-runtime==4.9.3
7 anyio==4.4.0
8 asgiref==3.8.1
9 asttokens==2.4.1
10 async-timeout==4.0.3
11 attrs==24.2.0
12 backoff==2.2.1
13 bcrypt==4.2.0
14 beautifulsoup4==4.12.3
15 boto3==1.35.68
16 botocore==1.35.68
17 build==1.2.1
```


运行后端

1 根据对应的 **requirements** 文件安装好 **Python** 依赖环境

```
requirements_mac.txt
requirements_ubun.txt
requirements_win.txt
```

2 在 **backend** 目录下运行 **uvicorn main:app --reload --port 8001**

The screenshot shows a code editor with a terminal window open. The terminal output shows the command `uvicorn main:app --reload --port 8001` being executed. The output includes an error message about address already in use, followed by information about the server starting and listening on `http://127.0.0.1:8001`. The terminal also shows several HTTP requests and responses, including `GET /documents?type=loaded` and `POST /load`. A notification popup is visible at the bottom, stating "Your application running on port 8001 is available. See all forwarded ports". The popup has buttons for "Open in Browser" and "Preview in Editor".

```
ERROR: [Errno 98] Address already in use
(20250214_RAG_Framework) huangj2@IHP-SPD-CF00015:~/Documents/RAG_P1_自制框架_250310/backend$ uvicorn main:app --reload --port 8001
INFO: Will watch for changes in these directories: ['/home/huangj2/Documents/RAG_P1_自制框架_250310/backend']
INFO: Uvicorn running on http://127.0.0.1:8001 (Press CTRL+C to quit)
INFO: Started reloader process [2829461] using WatchFiles
/mnt/external_disk/venv/20250214_RAG_Framework/lib/python3.10/site-packages/pydantic/_internal/_fields.py:161: UserWarning: Field
ected namespace "model_".

You may be able to resolve this warning by setting `model_config['protected_namespaces'] = ()`.
warnings.warn(
INFO: Started server process [2829463]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:50576 - "GET /documents?type=loaded HTTP/1.1" 200 OK
INFO: 127.0.0.1:50576 - "GET /documents?type=loaded HTTP/1.1" 200 OK
INFO: pikepdf._core:pikepdf C++ to Python logger bridge initialized
INFO: 127.0.0.1:50586 - "POST /load HTTP/1.1" 200 OK
INFO: 127.0.0.1:50586 - "GET /documents?type=loaded HTTP/1.1" 200 OK
INFO: 127.0.0.1:39294 - "POST /load HTTP/1.1" 200 OK
INFO: 127.0.0.1:39294 - "GET /documents?type=loaded HTTP/1.1" 200 OK

main:Total documents found: 14
127.0.0.1:32832 - "GET /list-embedded HTTP/1.1" 200 OK
pymilvus.orm.connections:Pass in the local path 03-vector-store/
127.0.0.1:59954 - "GET /collections/milvus HTTP/1.1" 200 OK
127.0.0.1:59984 - "GET /search-results HTTP/1.1" 200 OK
127.0.0.1:32828 - "GET /generation/models HTTP/1.1" 200 OK
pymilvus.orm.connections:Pass in the local path 03-vector-store/
127.0.0.1:59968 - "GET /collections?provider=milvus HTTP/1.1" 200 OK
127.0.0.1:39308 - "GET /providers HTTP/1.1" 200 OK
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pymilvus.orm.connections:Pass in the local path 03-vector-store/
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127.0.0.1:32828 - "GET /search-results HTTP/1.1" 200 OK
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pymilvus.orm.connections:Pass in the local path 03-vector-store/
127.0.0.1:59968 - "GET /collections?provider=milvus HTTP/1.1" 200 OK
```

Your application running on port 8001 is available. See all forwarded ports

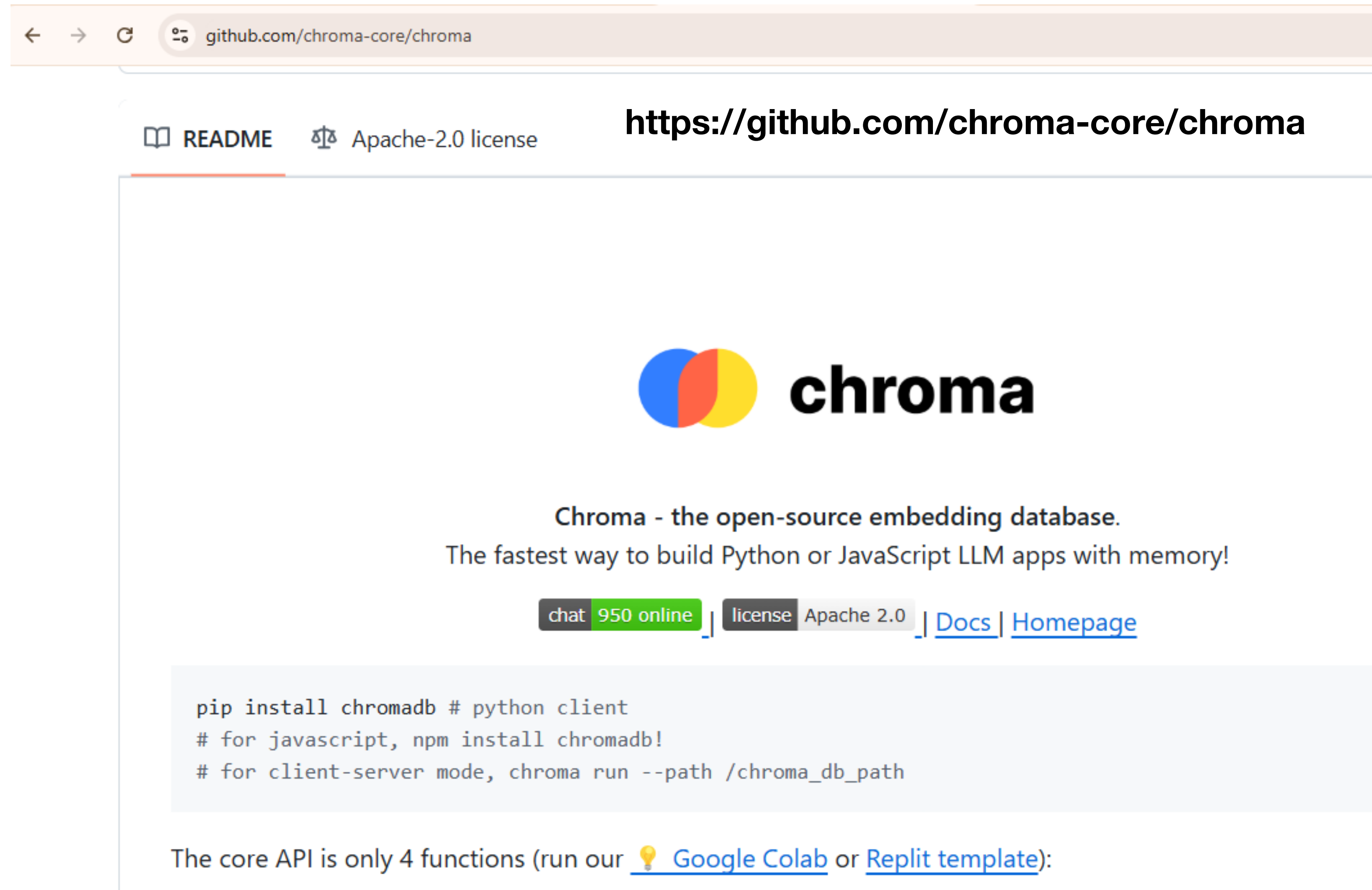
Open in Browser Preview in Editor

框架设计和流程说明（前端）

框架设计和流程说明（后端）

为框架添加新功能

新增加一个向量数据库：Chroma



从头开始构建框架

THANKS

 极客时间 | 训练营